The Eagle Has Wrapped: Practical Public-r Key Recovery against EagleSignV3

Martin Feussner
Selmer Center, University of Bergen
martin.feussner@uib.no

15 October 2026

This audit used the OpenAI model Daybreak Blue at the ultra reasoning level. OpenAI Codex performed AI-assisted attack-history research, paper review, cryptanalysis, implementation, experimentation, and technical writing. OpenAI Codex generated and selected this paper's title and deliberately chose "The Eagle Has Wrapped" as wordplay on "The Eagle Has Landed" and on the centered-modulus wrap that exposes the secret-key constraints. Separately tasked AI agents reviewed bounded parts of the argument and implementation. The primary full-parameter reproduction, separately tasked hostile adversarial review, prospectively sealed fresh-context independent reproduction, final exact-target history review, supplemental public-package replay, and independent Publication Review all passed. The Publication Reviewer returned `PASS / NO VETO`. No independent human verification was performed; it is optional for this audit.

Verified attack. Independent Publication Review returned `PASS / NO VETO`. The package, paper, reproducer, evidence pins, and release metadata satisfy the audit's publication gates.

Abstract

We describe a practical key-recovery and fresh-message forgery against the paper-level Ring Level 2 instantiation of EagleSignV3 in IACR ePrint 2024/1234. The recommended row sets $q = 59393$ and $\gamma_2 = 16384$, although the paper separately requires $q > 4\gamma_2$. Because this inequality is violated, adding the short product Dc can cross the centered representative boundary at $\pm(q-1)/2$ while satisfying the signer's printed low-bits test.

For an ordinary returned signature (r, z), the public verifier computes $v = Tz - Ac = p + Dc \pmod q$. Coordinates of v within $\beta = 16$ of a centered boundary have two possible pre-addition high bits. The public commitment r selects the correct combination. Every selected boundary flip is therefore a noiseless threshold inequality in one known signed 16-term convolution row of the ternary secret D. Accumulating these inequalities gives a small ternary integer program. Once D is recovered and public T is a ring unit, the other signing matrix follows publicly as $G = T^{-1}(A + D)$.

A controlled full-parameter replay retained all 40,000 outputs returned by the printed signing algorithm. It extracted 21,933 inequalities, including 1,811 wraps, and solved the resulting 21,933-by-1,024 ternary system with one SciPy/HiGHS worker in about 68 seconds and 213,404 KiB peak RSS. A prospectively sealed fresh-key reproduction independently retained 40,000 outputs, extracted 21,982 inequalities including 1,767 wraps, and recovered the exact signing key in a complete 296.09-second public attack pipeline using at most 221,272 KiB. Both candidates satisfied every inequality and the public relation before hidden-key comparison, then produced accepted signatures on unqueried messages with changed-message and altered-signature controls rejecting. The runs used, respectively, the Appendix's 32-byte and the main text's 64-byte completion of r. Each is one exact-row demonstration; neither estimates a minimum query count or an all-key success probability. No exact EagleSignV3 source or known-answer vector was located, so the claim is mathematical and paper-level rather than an implementation-conformance result.

1 Introduction

EagleSignV3 is a Fiat–Shamir-with-aborts lattice signature designed as a successor to EagleSign V1 and V2 [1]. Its public key is (ρ, T), where ρ expands to a public matrix A and

$$T = (A + D)G^{-1} \pmod q$$

for short secret matrices D and G. The signature changes the predecessors' response to $z = y + Gc$ and applies rejection tests to z and the low bits of $p + Dc$, where $p = Ty$. The paper presents this as the change that restores the zero-knowledge property after the attacks of Pulles and Tibouchi [2].

The Ring Level 2 parameter row does not satisfy a condition stated in the same paper. It chooses $q = 59393$ and $\gamma_2 = 2^{14}$, so

$$q = 59393 < 65536 = 4\gamma_2.$$

This matters because EagleSignV3 first reduces coefficients to centered representatives modulo q and then decomposes them modulo $2\gamma_2 = 32768$. A short addition near $\pm(q-1)/2$ can wrap through the centered modulus boundary and change the high part by two. The low part still lies well inside the signer's acceptance interval.

The hash value r commits to the high bits before Dc is added. The public signature and public key reveal the high bits after that addition. Near a boundary, testing the few possible wrap patterns against r recovers exact comparison outcomes for coefficients of Dc. Because c has Hamming weight 16 and D is ternary, each outcome is a sparse integer inequality in the 1,024 secret coefficients. Standard mixed-integer feasibility then recovers D and, for unit T, the equivalent signing key G.

We make no claim that boundary analysis, sparse linear inequalities, or mixed-integer solving is a new cryptanalytic technique. The result is their application to the exact EagleSignV3 Ring Level 2 equations and parameters, followed by a full key-recovery and ordinary-verifier fresh-message forgery. The final history result is a bounded search statement, not an absolute firstness or attack-technique novelty claim.

Contributions. We isolate the exact centered-wrap condition, prove that public r labels the resulting sparse inequalities, give a public-only recovery procedure, and report two complete one-key full-parameter executions under distinct r-length completions. We explicitly separate the paper-level mathematical result from byte-level completion choices and from the behavior of any unavailable author implementation.

2 Exact target and claim boundary

2.1 Pinned paper

The target is Hounkpevi, Djimnaibeye, Seck, and Sow's *EagleSignV3: A new secure variant of EagleSign signature over lattices*, IACR ePrint 2024/1234, revised 6 August 2024 [1]. The pinned 35-page PDF has SHA-256

`9b346bf2234299eadc72460a47c20dd71e6131fa19b3d21abdaaea01a8f061a2`.

No exact public EagleSignV3 implementation or known-answer vector was located. The paper says that an implementation can be deduced from EagleSignV2, but V2 is a predecessor with different signing equations and an explicit boundary guard. It is not a conformance oracle for this target. All executable claims in this report therefore concern an explicit reconstruction of the printed V3 mathematical scheme.

2.2 Frozen Ring Level 2 row

The attacked row is the ring variant at the paper's NIST security level 2:

Quantity	Frozen value
Ring	$\mathbb{Z}_q[X]/(X^{1024}+1)$
(k, l, n)	$(1, 1, 1024)$
q	59393
Challenge weight τ	16
Secret coefficient bound η	1
$\beta = l\eta\tau$	16
(γ_1, γ_2)	$(16384, 16384)$
Printed classical strong-unforgeability estimate	117 bits
Printed classical long-term key-recovery estimate	172 bits

Table 1: The exact paper-level parameter row used by the attack.

The paper defines centered representatives modulo odd q in $[-(q-1)/2, (q-1)/2]$, and centered representatives modulo even $2\gamma_2$ in $(-\gamma_2, \gamma_2]$. We use those conventions literally. Decomposing unsigned residues directly would describe a different scheme.

The paper's hash and byte descriptions conflict in several places. In particular, Section 3.3 gives a 512-bit range for hash G, making r 64 bytes, while Appendix A.4 says 256 bits, making r 32 bytes. The main signing algorithm and appendix also disagree on the length of λ. The primary replay used the Appendix's 32-byte r completion. The prospectively sealed independent reproduction and the proposed public reproducer use the main text's 64-byte r completion and Appendix A.4's 48-byte λ. Both profiles fix all remaining byte choices before execution. The wrap argument depends only on the printed ring equations, centered decomposition, challenge distribution, and public commitment to the pre-addition high bits; its success under both r lengths is therefore paper-level evidence across the conflict, not evidence for a canonical wire format.

2.3 Output semantics

The claim consumes every value returned after Algorithm 4's two printed rejection checks. In the primary run, 1,767 of 40,000 returned values—about 4.4%—were rejected when subsequently submitted to the ordinary verifier. That mismatch is another consequence of the broken wrap invariant. The returned values still expose public (r, z) and therefore valid wrap labels for the recovery system.

An implementation that adds an extra ordinary self-verification test and resamples failed outputs defines a different signing procedure. We do not claim that the present 40,000-output experiment transfers unchanged to that modified procedure. This boundary is essential to the result.

3 EagleSignV3 equations

Let $R_q = \mathbb{Z}_q[X]/(X^n + 1)$. At the attacked ring row all matrices are single ring elements. Key generation samples ternary $D, G \in R_q$, with G invertible, expands ρ to uniform $A \in R_q$, and publishes

$$T = (A + D)G^{-1} \pmod q.$$

To sign a message M, the signer computes a message digest μ, samples $y \in S_{\gamma_1}$, and forms

$$p = Ty, \qquad r = G_{\text{hash}}(\mathsf{HighBits}_q(p)), \qquad c = H(\mu, r), \qquad z = y + Gc.$$

The challenge c has exactly $\tau = 16$ coefficients in $\{-1, +1\}$ and all other coefficients zero. The signer returns (r, z) after checking

$$\|z\|_\infty < \gamma_1 - \beta, \qquad \|\mathsf{LowBits}_q(p + Dc)\|_\infty < \gamma_2 - \beta.$$

The verifier recomputes $c = H(\mu, r)$ and

$$v = Tz - Ac.$$

Using $TG = A + D$ and $z = y + Gc$ gives the central public identity

$$v = Ty + (TG - A)c = p + Dc \pmod q. \tag{1}$$

It accepts when z is short and $c = H(\mu, G_{\text{hash}}(\mathsf{HighBits}_q(v)))$.

The paper separately instructs designers to choose $q > 4\max(\gamma_1, \gamma_2)$. That condition would place enough room between the decomposition scale and the centered q boundary for the intended high-bit stability argument. Table 1 violates it.

4 Public wrap labels

Write

$$h = (q-1)/2 = 29696, \qquad B = 2\gamma_2 = 32768, \qquad s = (Dc)_j.$$

Because D is ternary and c has weight 16, $|s| \le 16 = \beta$. A coordinate of the public v can have crossed the centered boundary only when its representative lies within 16 units of $-h$ or $+h$.

Lemma 1 (Lower boundary)**.** *Suppose $v_j = -h + a$ with $0 \le a < \beta$. The hidden pre-addition coefficient p_j lies on the positive side of the centered boundary exactly when $s > a$.*

Proof. Equation (1) gives $p_j = v_j - s \pmod q$. Its centered representative wraps from the displayed negative value to the positive side exactly when $v_j - s < -h$. Substituting $v_j = -h + a$ yields $s > a$. □

Lemma 2 (Upper boundary)**.** *Suppose $v_j = h - a$ with $0 \le a < \beta$. The hidden pre-addition coefficient p_j lies on the negative side of the centered boundary exactly when $s < -a$.*

Proof. Here $p_j = v_j - s$ crosses above h exactly when $h - a - s > h$, or $s < -a$. □

For a lower-edge coordinate, $\mathsf{HighBits}_q(v_j) = -1$ while the wrapped alternative for $\mathsf{HighBits}_q(p_j)$ is $+1$. For an upper edge the public value is $+1$ and the wrapped alternative is -1. If one signature contains K edge coordinates, the attacker enumerates their 2^K possible toggles, hashes each complete high-bit vector, and compares with public r. A unique match identifies every wrap bit. In the primary run, K ranged from zero to five; all 40,000 outputs had a unique match, and only 69,252 candidate vectors were tested in total.

5 From labels to the signing key

Let the nonzero challenge coefficients be $(c_b)_{b \in S}$ with $|S| = 16$. Negacyclic multiplication gives

$$(Dc)_j = \sum_{b \in S} \epsilon_{j,b} c_b D_{j-b \bmod n},$$

where $\epsilon_{j,b} = +1$ before the $X^n = -1$ fold and -1 after it. Every coefficient is therefore a known signed sum of 16 unknowns $D_i \in \{-1, 0, 1\}$.

The recovered lower-edge label supplies either

$$(Dc)_j \ge a + 1 \quad \text{or} \quad (Dc)_j \le a,$$

and the upper-edge label supplies either

$$(Dc)_j \le -a - 1 \quad \text{or} \quad (Dc)_j \ge -a.$$

These are exact integer inequalities, with no statistical noise term. Pooling them over returned signatures gives a ternary feasibility problem in 1,024 variables.

After solving for a candidate D, the attacker performs two public checks. First, every collected inequality must hold exactly. Second, when public T is a unit in R_q, compute

$$G = T^{-1}(A + D) \pmod q. \tag{2}$$

The centered coefficients of G must be ternary, G must be invertible, and $TG = A + D$ must hold. These checks select a signing-equivalent pair using public data alone; comparison with the generated hidden key is unnecessary for the attack and is postponed until after the forgery gate.

For the paper-level Ring Level 2 distribution, the estimated probability that T is a unit is

$$(1 - 1/q)^n \approx 0.9829.$$

The direct bridge in Equation (2) is explicitly conditional on this public property. A nonunit T does not invalidate the inequality recovery, but deriving a complete signer would need another method and is not claimed here.

Theorem 1 (Conditional signing compromise)**.** *For the paper-level Ring Level 2 equations, suppose returned signatures give a ternary D satisfying all public wrap inequalities, public T is a unit, and Equation (2) yields a ternary invertible G satisfying $TG = A + D$. Then (D, G) is an equivalent signing key and can run the printed signing equations on a fresh message.*

Proof. The public checks establish the exact key-generation relation for the same A, T. Substituting the recovered pair into signing yields $Tz - Ac = Ty + (TG - A)c = Ty + Dc$, exactly the correctness identity used by ordinary verification. Rejection sampling can be repeated until the printed bounds and ordinary verification accept. No hidden value from the original key is needed. □

6 Full-parameter evidence

6.1 Primary reproduction: Appendix 32-byte r completion

The controlled primary replay fixed one Ring Level 2 key and retained exactly 40,000 outputs of the printed signing algorithm. Table 2 summarizes the stable result.

Measurement	Result
Printed signer outputs	40,000
Public edge inequalities	21,933
Centered-boundary wraps	1,811
Ordinary-verifier-invalid returned outputs	1,767 (4.4175%)
Hash candidates tested	69,252
Ternary variables / nonzeros	1,024 / 350,928
SciPy/HiGHS branch-and-bound nodes	1
SciPy/HiGHS solver / command wall time	65.67 s / 68.08 s
SciPy/HiGHS peak RSS / workers	213,404 KiB / 1
Fresh-message ordinary verification	accepted
Changed-message / altered-r controls	rejected / rejected

Table 2: Primary paper-level Ring Level 2 reproduction.

Collection took 598.77 seconds. The regenerated constraint dataset has SHA-256

`a15776674731b97cca2c469ab4d43512c966d6d4636e5e59344ffd8d38ffb97c`.

SciPy/HiGHS returned a ternary candidate satisfying all 21,933 inequalities. Before any hidden-key comparison, the implementation derived ternary invertible G, checked $TG = A + D$, and signed the never-queried message

`eaglesignv3-highs-continuation-fresh-message-never-queried`.

The ordinary verifier accepted that signature. Reusing it on a changed message failed, and flipping one bit of r while retaining z also failed. Only after these public checks passed did a post-hoc comparison confirm exact recovery of the generated D and G.

The fresh-forgery and recovered-key artifacts have SHA-256 values

`1e90009b4eb0d9e9cb0103c6015ee7b3a76bb47a6a4334b0ac703fa79eeb5822`,
`e0e0fa88ff3bef951687344d17731018495cb958bf68e50d65f7cff3d0bf4eb1`.

The fixed count 40,000 is one demonstrated working point. We did not optimize the number of outputs and make no minimum-query claim. The experiment used one fixed key and therefore does not measure a distribution-wide success rate.

6.2 Hostile adversarial review

A separately tasked reviewer worked with the explicit objective of disproving the attack and returned `PASS_WITH_LIMITATIONS / NO_VETO`. The review covered the exact Ring Level 2 arithmetic, both strict wrap inequalities, public-r label selection, the sparse negacyclic convolution rows, ternary solver bounds, public derivation of G, fresh-message signing, ordinary verification, and the primary endpoint. It found no hidden-key input or ground-truth label in candidate selection. The separate verifier shares the audited public model code, which remains a disclosed evidentiary limitation.

The review preserved the material boundaries stated here: paper-level Ring Level 2 only; one fixed primary key; direct signing conditional on unit T; 40,000 as a working point rather than a minimum; use of every output returned by printed Algorithm 4; explicit completion of conflicting byte details; and no author-implementation claim. Its handoff, full review, verdict, and sealed artifact manifest have SHA-256 values

`e8c51f43aff9c1a57bd44a619bb718715ee2d340c9e29c01b46228a524f881cb`,

af6fec885f924f5df83e202fec9b05106c897fe53d5edb7a7a1f1e2218e218dc,
e08c94ef6da1dcbc3a66f46cd231054a345b056ab30b524e6f09fa36638bb420,
4e1bcf3d6607514189b4b1362805e8b361c5433cc774b54af405547d049daad7.

This gate authorizes the sealed independent reproduction; it does not itself count or publish the attack.

6.3 Fresh-context independent reproduction: main-text 64-byte r completion

A fresh-context implementation and run were prospectively sealed before the target seed was created. The seal covered the hypothesis, fixed completion profile, exact commands, C++20 source, SciPy/HiGHS solver wrapper, and all six executables. The fresh 32-byte kernel-CSPRNG seed was generated only after that seal. Key attempt zero had unit public T and was the only generated key.

The independent completion follows Section 3.3's 512-bit definition of hash G, so r is 64 bytes. It follows Appendix A.4 for 48-byte λ and 32-byte ρ and tr, uses 48-byte μ, and samples uniformly from the inclusive set $[-\gamma_1, \gamma_1]$. This differs deliberately from the primary replay's Appendix-based 32-byte r completion while preserving the same printed Ring Level 2 equations and attack mechanism.

The run retained every one of 40,000 outputs returned after Algorithm 4's two printed rejection tests. Of those, 38,273 passed and 1,727 failed the ordinary verifier; no verifier filter was applied. Public edge enumeration produced 21,982 exact ternary inequalities, including 1,767 edge flips, from 69,301 candidate high-bit vectors. The largest single transcript had six edge coordinates. SciPy 1.18.1 `optimize.milp`, using bundled HiGHS 1.12.0, solved the 1,024-variable, 351,712-nonzero system in 141.57 solver seconds at one node with zero gap and zero checked row violation.

Before hidden-key access, the implementation checked ternarity, every inequality, unit G, and $TG = A + D$, then signed the never-queried message

`EagleSignV3 independent fresh forgery 2026-10-15 / ESV3-IR-001`.

The unchanged ordinary verifier accepted the untouched forgery and rejected both a changed message and an altered signature. Only afterward did a separate process compare against ground truth and confirm exact D and G matches.

The oracle took 124.51 seconds, constraint extraction 12.83 seconds, the SciPy/HiGHS command 144.56 seconds, recovery and fresh signing 12.82 seconds, and public verification 1.37 seconds. The complete public pipeline therefore took 296.09 seconds, excluding post-hoc comparison, with one worker, one-CPU solver affinity, a 4 GiB virtual-memory ceiling, and 221,272 KiB peak RSS.

The prospective seal, run artifact manifest, solver report, recovered key, fresh forgery, and terminal handoff have SHA-256 values

2e8144384c31f59a3dd4d2d196f2d51da885c8bd8fde25e0d7713440e2b67934,
19da75184dde5c2697d8f796bdd3e33a0b09c7700318151ff4c0a9e42613de8a,
4e66d7a7c57bdc073088e5c63320c31fb00b0b4b448a5b2514dc639ac89dedc9,
4dc18591edf542ccd5ea1f5ef13cb01f00973694b9a1e62510bac4a193acc737,
aad6f1610b96b5d5b5d10e961d334b1a240fc7f7d4e5066ae7a42fc7316345c5,
bf51fc87cc1c7f2060c1e30caea121d4f57183e2496424a3646f00672e0473b0.

6.4 Publication Review and supplemental package replay

Independent Publication Review returned `PASS / NO VETO`. As supplemental release-readiness validation, the complete public-package path was also replayed on a new fresh unit-T key. It retained all 40,000 printed signer outputs: 38,219 passed ordinary verification and 1,781 failed it, with no verifier filtering. The replay produced 22,083 exact inequalities including 1,821 wraps, recovered exact D and G, checked the public relation, and produced an accepted fresh-message forgery. The unchanged verifier rejected the changed message and altered signature. The one-worker solve took exactly 83.31292366300477 solver-wrapper seconds and 84.10 command-wall seconds, with 243,116 KiB peak RSS.

This package replay is supplemental validation of the release path. It does not replace the prospectively sealed independent reproduction, enlarge the claim beyond the primary and independent two-key trial scope, or change the reported two trials and two successes. The replay review, machine-readable report, retained output manifest, and review artifact manifest have SHA-256 values

ee6fefb703b8be1c50f5d2804c57f5a62bc1f7f377f1bf8ca0f7aef3769b7761,
b1549fd2456b8f68d8393ddca1cf23e1ecd2cb43fc12cddb4337832796583e9a,
b969b537a02e87439a2d07d25a44592e23fb5a187526bbcec9ff6ba8f4427fad,
8437db885f97a5021353cff69bef8b1cf7604ddbc200feb458196938da369837.

The replay input was the pre-finalization source candidate, pinned by root manifest `9c8fecf3ba1ea16df8ebd60c52d117a014b0939ac1bb9c85a9fea511a25c93d5` and nested reproducer manifest `d5b48642fe37e2a043af4b6e9a79219452e9a73d4583c31d4c8a297579e249ac`. These are old source-candidate pins, not hashes of this final package.

7 Reproduction architecture

The public reproducer is an independently written C++20 paper-level implementation. It uses separate executables for key generation, oracle collection, public recovery, ordinary verification, and post-hoc ground-truth comparison. The recovery executable receives only the public key and the returned (M, r, z) records. It writes an LP-format integer program. A sealed Python wrapper parses that file into a sparse matrix and invokes SciPy 1.18.1 `optimize.milp` with bundled HiGHS 1.12.0, integral ternary bounds, one thread, and one-CPU affinity. The wrapper independently validates every inequality before emitting its candidate. Recovery then derives G publicly and creates the fresh signature. The verifier separately checks freshness, the public relation, ordinary acceptance, changed-message rejection, and an altered-signature rejection before ground truth is consulted.

The package offers a structural smoke test and a full fresh-key run:

```
./reproduce.sh --smoke
./reproduce.sh --full
```

Full mode uses one worker, a 4 GiB virtual-memory ceiling, 40,000 returned outputs, and the sealed 21,600-second SciPy/HiGHS time limit. Generated secret material and the large transcript remain in a temporary directory and are deleted after the final post-hoc comparison.

Because the paper has no exact V3 source and contains conflicting byte details, the reproducer records its choices: a 64-byte r following the main text, 48-byte λ, 32-byte ρ and tr, 48-byte μ, uniform inclusive $[-\gamma_1, \gamma_1]$ sampling, canonical little-endian public coefficients, and a documented high-bit encoding. These choices preserve the printed equations and distributions but do not define a canonical V3 wire format.

8 Attack history and attribution

The final post-reproduction history review returned `PASS`, with moderate confidence and no history veto. Its approved bounded statement is:

> No prior public substantive attack against EagleSignV3 (IACR ePrint 2024/1234) was located in the searches performed as of 15 October 2026.

This statement cannot exclude private, removed, unpublished, or unindexed work, and search and citation indexes can lag.

The direct-lineage V1/V2 cryptanalysis is documented in IACR ePrint 2024/1137 [2]. Pulles and Tibouchi previously broke EagleSign V1 and V2 using response statistics and ideal structure. Those schemes use different response relations, and EagleSign V2 includes an explicit centered-modulus-boundary rejection that is absent from the EagleSignV3 specification studied here. We located no earlier public disclosure applying the centered-modulus-wrap, public-commitment-label route to EagleSignV3. This is a bounded search statement; we make no absolute firstness or attack-technique novelty claim.

Prabowo and Tan analyze generic Schnorr-type signatures with aborts and give a statistical attack on the original EagleSign [3]. Their work provides context for response statistics but does not attack the exact V3 scheme or analyze the centered-wrap public-r route. Pellet-Mary and Seck study generic properties of HighBits and LowBits and their applications [4]. The retrieved abstract, keywords, visible references, and indexed search surfaces disclosed no EagleSign attack; we treat that work as adjacent technique context rather than an exact-target attack.

The canonical history handoff, review handoff, machine-readable verdict, query ledger, source ledger, and review artifact manifest have SHA-256 values

`2a1cbf638453500b5749cbc040ece0d03abb453e0bb95f7ccd7f4d4f2df82faa`,
`53b5c8f3ca4726764b9e0db6df5f6c458f25d988b693968044245a01bfe31d62`,
`b28391fda40777c64919a61035a33f8135eccb60a2f635cc7e78d793e2093236`,
`43b6eb77e399f8119fc88b162db57ff80d2d8dbf4a5ab127bfcfe4af700256c2`,
`f71cd94c1e1e6a9a3af690e39c63074e61aaf98aea7e1bd16705951b5c2f2f48`,
`d50ad9e9bebf5a1855c89f960592e131f19fce2b87d8bb5e4514f567d217454d`.

The attack technique itself is not claimed as original. The contribution is the exact application of standard boundary reasoning and integer feasibility to the published EagleSignV3 Ring Level 2 row, with a complete signing-key bridge and fresh-message forgery.

9 Limitations and repair boundary

The following qualifications are part of the claim:

- The result covers only the exact paper-level Ring Level 2 row in Table 1. It does not claim the module rows, Level 5 rows, or any unpublished implementation.
- No exact V3 source or known-answer vector was located. Conflicting hash lengths, seed lengths, and encodings required explicit completions; there is no author-source or wire-format conformance claim.
- The direct signing-key bridge applies when public T is a unit, estimated at about 98.29% for this paper-level row and detectable from the public key.
- The primary and independent evidence are each one-key executions. They use different keys, implementations, and r-length completions, but together remain a two-key record rather than a many-key success-rate study.
- Forty thousand outputs were sufficient in both reported runs; they are not asserted to be necessary or minimal.
- The attack uses every output of the printed signer, including the roughly 4.4% that ordinary verification rejects. An added self-verification-and-resampling loop is outside the claim.
- The final history PASS has moderate confidence and is a dated bounded search result. It cannot establish absolute priority or exclude private, deleted, unpublished, or unindexed work.

The paper's own condition $q > 4\max(\gamma_1, \gamma_2)$ excludes the reported parameter mismatch. Enforcing that condition in a revised row, or restoring a correct centered-boundary guard before returning signatures, are natural starting points for repair. Either change alters the analyzed scheme or its parameters and requires fresh correctness, rejection-rate, security, and implementation analysis. Adding only a final self-verification loop blocks the precise output stream used here, but is likewise a modified signer and is not by itself a security proof.

10 Conclusion

At EagleSignV3 Ring Level 2, the printed modulus is too small for the paper's own decomposition condition. Public r resolves the resulting centered-wrap choices and converts ordinary signer outputs into exact sparse inequalities in the ternary secret D. One full-parameter primary run solved 21,933 such inequalities under the Appendix's 32-byte r completion. A prospectively sealed fresh-key reproduction independently solved 21,982 inequalities under the main text's 64-byte r completion. Each derived G from the public relation and produced an accepted fresh-message signature with no hidden-key input to recovery. The result is a practical paper-level signing compromise on two demonstrated keys, subject to the explicit unit-T, output-semantics, and specification-completion limits above. Independent Publication Review returned `PASS / NO VETO`, and the supplemental public-package replay confirmed the complete release path without changing the two-key claim scope.

Acknowledgements

The computations were performed on the Norwegian Research and Education Cloud (NREC), using resources provided by the University of Bergen and the University of Oslo.

References

[1] A. C. Hounkpevi, S. Djimnaibeye, M. Seck, and D. Sow, "EagleSignV3: A new secure variant of EagleSign signature over lattices," IACR Cryptology ePrint Archive, Report 2024/1234, revised 6 August 2024. https://eprint.iacr.org/2024/1234

[2] L. N. Pulles and M. Tibouchi, "Cryptanalysis of EagleSign," IACR Cryptology ePrint Archive, Report 2024/1137, 2024. https://eprint.iacr.org/2024/1137

[3] T. F. Prabowo and C. H. Tan, "Concrete Analysis of Schnorr-type Signatures with Aborts," IACR Cryptology ePrint Archive, Report 2024/1256, 2024. https://eprint.iacr.org/2024/1256

[4] A. Pellet-Mary and M. Seck, "On the Properties of HighBits and LowBits Functions and their Applications," IACR Cryptology ePrint Archive, Report 2026/895, 2026. https://eprint.iacr.org/2026/895

[5] Q. Huangfu and J. A. J. Hall, "Parallelizing the dual revised simplex method," *Mathematical Programming Computation*, vol. 10, pp. 119–142, 2018.